

Xiaotian Zhou

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EDUCATION

The Pennsylvania State University

Ph.D. Candidate, Computer Science and Engineering

Advisors, Prof. Syed Rafiul Hussain & Prof. G. Gary Tan

08/2025–Present

University Park, PA

The Pennsylvania State University

Master of Science, Computer Science and Engineering

Advisor, Prof. Syed Rafiul Hussain

08/2023–05/2025

University Park, PA

ShanghaiTech University

Bachelor of Engineering, Computer Science and Technology

Visiting Student, University of California, Berkeley (08/2021–05/2022)

Thesis, Mitigating Adversarial Attacks against Synthetic Audio Detection Systems

08/2019–06/2023

Shanghai, China

RESEARCH INTERESTS

Formal Verification, Computer Security, Programming Languages, Proof Assistants, Cryptography

RESEARCH EXPERIENCE

The Pennsylvania State University

Research Assistant

09/2023–Present

University Park, PA

- **Verified ASN.1 UPER Parser**: Built a generator of verified UPER parsers in Rocq with bit-level round-trip proofs, and used it as a differential-testing oracle to find bugs in 11 widely used ASN.1 parsers [2].
- **Rust Security**: Designing and implementing a sanitizer for Rust aliasing undefined behavior.

Seagate Technology

Research Intern - Security and Cryptography

05/2026–08/2026

Shakopee, MN

- Developed a computational security proof of HPKE (RFC 9180) as used in OCP L.O.C.K., the Open Compute Project's key-management specification for self-encrypting drives, using SSProve and Hax.

PUBLICATIONS

- [1] Kai Tu, Tianchang Yang, Xiaotian Zhou, Ali Ranjbar, Abdullah Al Ishtiaq, Tianwei Wu, Yilu Dong, and Syed Rafiul Hussain. **Octopus: Target-Aware Adaptive Field-level Fuzzing for Cellular ASN.1 Protocols**. IEEE Symposium on Security and Privacy (S&P), 2027.
- [2] Xiaotian Zhou, Kai Tu, Ali Ranjbar, Yilu Dong, Gang Tan, and Syed Rafiul Hussain. **VUPER: Verified ASN.1 UPER Parser**. The ACM Conference on Computer and Communications Security (CCS), 2026.

TEACHING EXPERIENCE

The Pennsylvania State University

08/2025–12/2025

Teaching Assistant - Computer Security

University Park, PA

- Designed and graded programming assignments and quiz questions for a class of 70+ students.
- Held weekly office hours to guide students on Cryptography and Computer Security concepts.

ShanghaiTech University

09/2020–06/2021, 01/2023–06/2023

Teaching Assistant - Mathematical Analysis I & II

Shanghai, China

- Conducted weekly recitation sessions for 30+ students.
- Graded assignments and exams on topics like Calculus.

PROFESSIONAL SERVICE

Artifact Evaluation Committee: CCS '26

External Reviewer: FutureG '26

AWARDS

- OPLSS Fellowship (\$1,780) *Oregon Programming Languages Summer School 2024*
- Exchange Student Program Scholarship *ShanghaiTech University 2022*
- Merit Student *ShanghaiTech University 2019–2020*

SECURITY FINDINGS

- **Unisoc:** Discovered 8 High-severity CVEs with Octopus [1], such as CVE-2025-71256, CVE-2026-21549, and CVE-2026-21550.
- **Apple:** Discovered a baseband vulnerability with Octopus [1], acknowledged in iOS 27 security notes.
- **asn1c:** Discovered CVE-2025-55398 (Critical) with VUPER [2].
- **Rocq Prover:** Discovered a critical soundness bug permitting a proof of **False**.

SKILLS

- **Relevant Coursework:** Computer Security, Cryptography, Parallel Programs & Processors, Machine Learning, Operating Systems, Compiler Construction, Securing Embedded Systems
- **Programming Languages:** C, C++, Python, Rocq Prover (Coq), OCaml, Rust
- **Software:** Docker, Git, L^AT_EX, Linux, PyTorch, NumPy, Ghidra, AFL++, LibAFL